



Statistics Practice Questions

Table 1 below shows the heights of 20 randomly selected U.S. men, age 20-29.²

Subject number	Height (inches)
1	72.2
2	71
3	72.8
4	63.8
5	68.1
6	69.4
7	74.2
8	75.8
9	66.3
10	66.5
11	69.2
12	66.7
13	69.4
14	75.1
15	72.3
16	68.5
17	68.7
18	68
19	68.5
20	68.6

Table 1: Measured heights of 20 randomly selected U.S. males, age 20-29.

- 1) Calculate the mean and the standard deviation of the heights. You may use a calculator, but if your calculator has a built-in functions for mean and standard deviation, do not use them.

² All data in this set of questions are modified from the following source: National Health Statistics Reports, Number 10, October 22, 2008. <http://www.cdc.gov/nchs/data/nhsr/nhsr010.pdf>

2) Calculate the standard error.

3) Plot a histogram of the data.

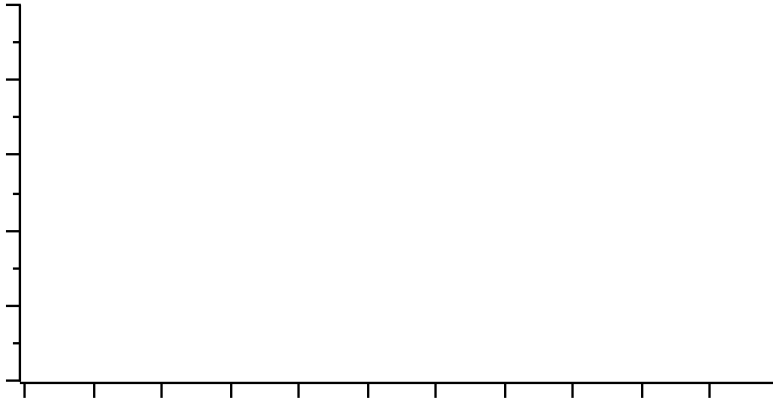


Figure 1 shows the heights sampled from two other populations, U.S. women age 20-29 and U.S. women age 60-69.

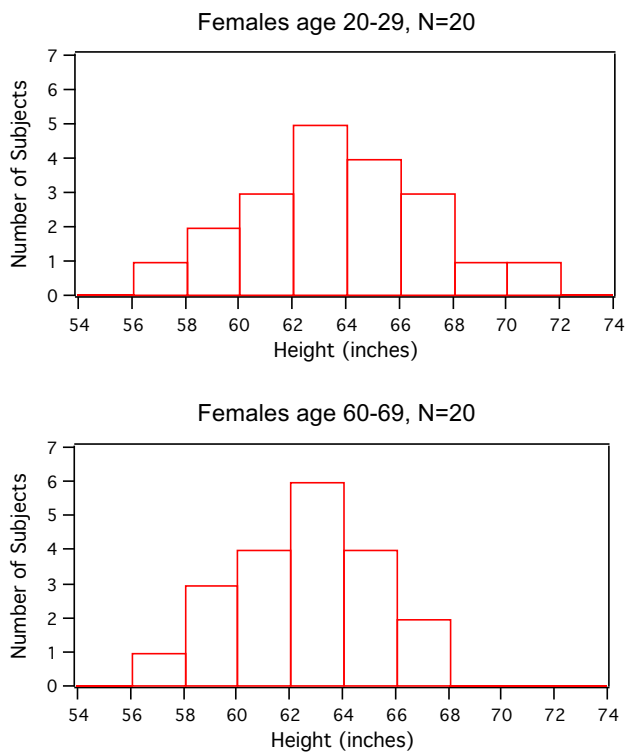
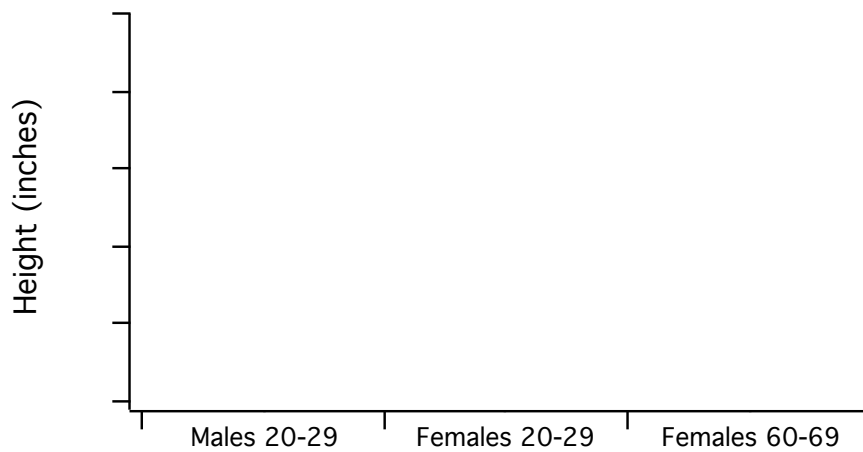


Figure 1: Heights sampled from two populations of U.S. females.

4) Estimate the mean and standard deviation from the graph for both samples in Figure 1 (you should not need a calculator for this).

5) From your estimates, calculate the standard error of the mean for both samples.

6) Plot a bar graph showing the means and standard errors of all 3 populations (males in their 20s, females in their 20s, females in their 60s)



7) Is there a statistically significant difference between the heights of females in their 20s and males in their 20s? Between females in their 20s and females in their 60s? Explain your reasoning.

Figure 2 shows heights collected from the same 3 populations (males in their 20s, females in their 20s, females in their 60s) but with sample sizes of 808, 1061, and 691 respectively.

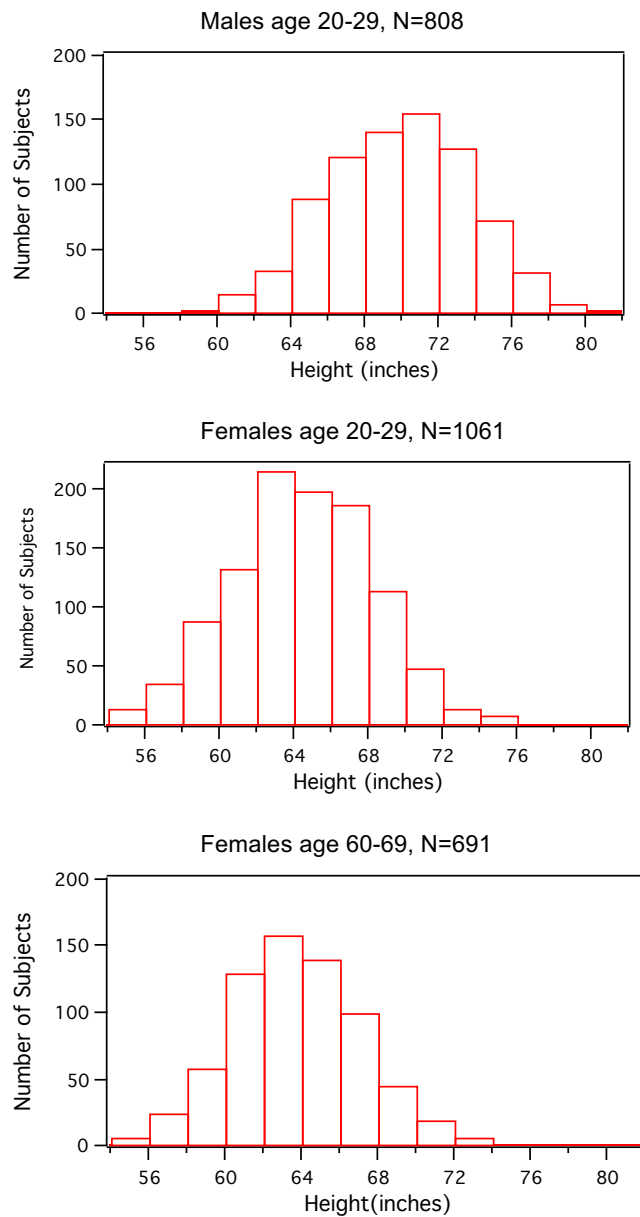
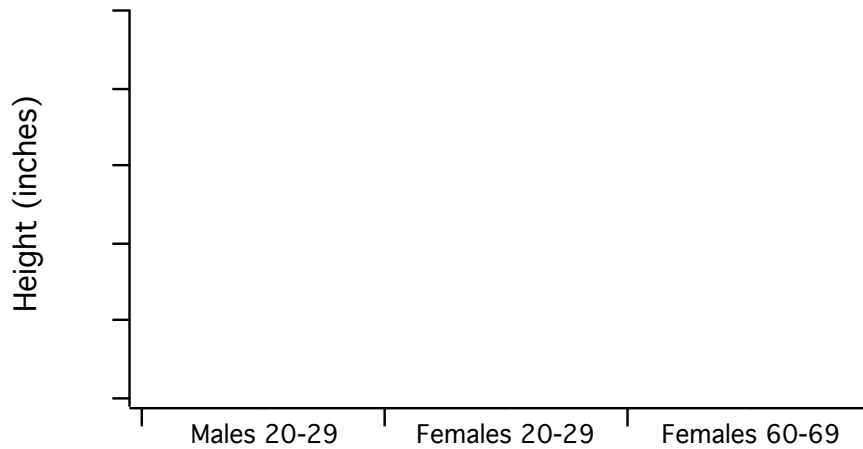


Figure 2: Measured heights sampled from three populations.

- 8) From the graphs, estimate the means and standard deviations for each population, and calculate the standard error.

- 9) Make a bar graph showing the means and standard errors for all 3 populations.



- 10) Is there a statistically significant difference between the heights of females in their 20s and males in their 20s? Between females in their 20s and females in their 60s? Explain your reasoning.

- 11) What effect does increasing the sample size have on the standard error? What effect does increasing the sample size have on the mean and standard deviation?



Calculating with Units Practice Questions

Useful unit conversions:

Time

1 minute = 60 seconds

1 hour = 60 minutes

1 day = 24 hours

1 year = 365 days (non-leap year)

Distance

1 mile = 1.61 km

1 km = 1000 m = 10^3 m

1 inch = 2.54 cm

1 ft = 12 inches

1 mile = 5280 ft

1 light year = 9.5×10^{15} m

Weight

1 kg = 2.2 pounds (lbs)

1 lb = 16 ounces

1 ounce = 28.3 g

Volume

1 cup = 8 oz (fluid ounces)

1 pint = 2 cups

1 quart = 2 pints

1 gallon = 4 quarts

1 gallon = 3.87 liters (L)

Part 1: Practice with fractions:

1)

$$\frac{2}{3} \cdot \frac{7}{5} = ?$$

2)

$$\frac{2/3}{5} = ?$$

3)

$$\frac{2}{3/5} = ?$$

4)

$$\frac{a/b}{c} = ?$$

5)

$$\frac{d}{e/f} = ?$$

6)

$$\frac{a/b}{c/d} = ?$$

Part 2: Time

What is:

- 1) 5 days, in hours?
- 2) 1 day, in seconds?
- 3) 2.5 hours, in seconds?
- 4) 10 minutes, in days?
- 5) 1 year, in minutes?

Part 3: Distance

What is:

- 1) 26.2 miles in km?
- 2) 100 meters in feet?
- 3) 5,000 miles in inches?
- 4) 5,000 miles in light years?
- 5) 5'2" (5 feet, 2 inches) in meters?

Part 4: Speed

What is:

- 1) 30 m/s in ft/s?
- 2) 600 miles per hour (mph) in miles per second?
- 3) 65 mph in m/s?
- 4) 3×10^8 m/s in miles per hour (mph)?
- 5) 3×10^8 m/s in light years per day?

Part 5: Area and Volume

- 1) Do units of ft^2 describe length, area, or volume?
- 2) What is 500 ft^2 in in^2 ?
- 3) Do units of m^3 describe length, area, or volume?
- 4) What is 55 m^3 in in^3 ?

- 5) A football field is 360 ft by 160 ft, including end zones. What is its area in cm^2 ? In m^2 ? In miles^2 ?

Part 6: Back of the Envelope Units

- 1) If your heart beats an average of 60 times per minute, how many times does it beat in a week?
In a year?
- 2) If 20 NYC street blocks equal a mile (not avenue blocks), how far is Times Square from Butler Library in miles? In km?
- 3) If Joe's sells 30 cups of coffee in an hour, then how many cups of coffee do they sell in 4 hours? How much coffee is that in gallons? In liters? Assume a typical cup of coffee is 12 oz. (You will need to look up conversions for fluid ounces (oz.), gallons, and liters.)
- 4) If a golden retriever eats 3 cups of food per day, a cup of dog food weighs about 100 grams and a 33 lb bag of dog food costs \$40, then how much money does it cost to feed the dog each year?
- 5) If the average piano tuner works about 255 days per year for 8 hours a day, and one piano takes about 2 hours to tune, and most piano owners have their pianos tuned about once per year, and if we can expect a population to have about 1 piano per 100 people, and New York City is home to 8 million people, then how many piano tuners are there in New York City?



Sense of Scale Practice Questions

Part 1: Practice with scientific notation

1) $123000 = 1.23 \times 10^7$

2) $602 \times 10^{21} = 6.02 \times 10^7$

3) $0.00098 = 9.8 \times 10^7$

4) $0.0022 \times 10^6 = 2.2 \times 10^7$

5) $160 \times 10^{-21} = 1.60 \times 10^7$

6) $0.000663 \times 10^{-30} = 6.63 \times 10^7$

7)

$$\frac{2}{3 \times 10^{-15}} = ?$$

8)

$$\frac{2 \times 10^{-15}}{3} = ?$$

9)

$$\frac{2 \times 10^{21}}{3 \times 10^3} = ?$$

10)

$$\frac{7 \times 10^{21}}{.2 \times 10^{-3}} = ?$$

Part 2: Orders of magnitude

Give an order of magnitude estimate for the following. Do as many as you can from memory, and try to guess those you don't know.

- 1) Diameter of a proton
- 2) Diameter of a hydrogen atom
- 3) Thickness of a piece of paper
- 4) Height of a skyscraper
- 5) Distance to the Moon
- 6) Distance to the Sun
- 7) Distance to the closest star outside of our Solar system
- 8) Distance to the center of the Milky Way galaxy (our galaxy)
- 9) Diameter of the observable Universe
- 10) Number of years since the last ice age
- 11) Years since *Tyrannosaurus rex* went extinct
- 12) Age of the Earth
- 13) Age of the Universe

Part 3: Relative magnitude

How many times greater is (A) compared to (B)? How many orders of magnitude larger is (A) compared to (B)?

- 1) A: Height of a person
B: Diameter of a proton
- 2) A: Distance to the closest star outside of our solar system
B: Height of a person
- 3) A: Thickness of a piece of paper
B: Diameter of a hydrogen atom
- 4) A: Distance to the Moon

B: Height of a skyscraper

5) A: Size of the observable Universe

B: Diameter of a proton

6) A: Age of the Universe

B: Age of the Earth

7) A: Age of the Earth

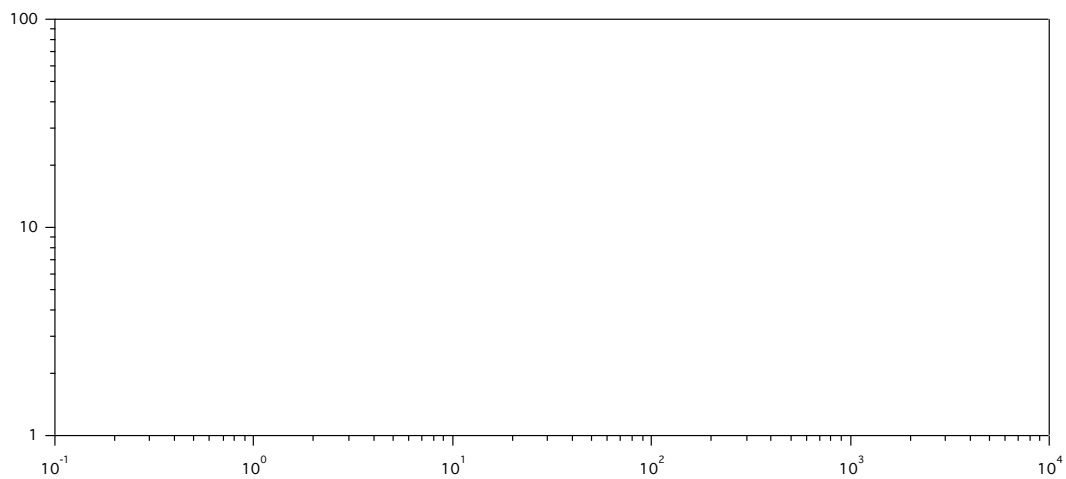
B: Years since *T. rex* walked the Earth

8) A: Years since *T. rex* walked the Earth

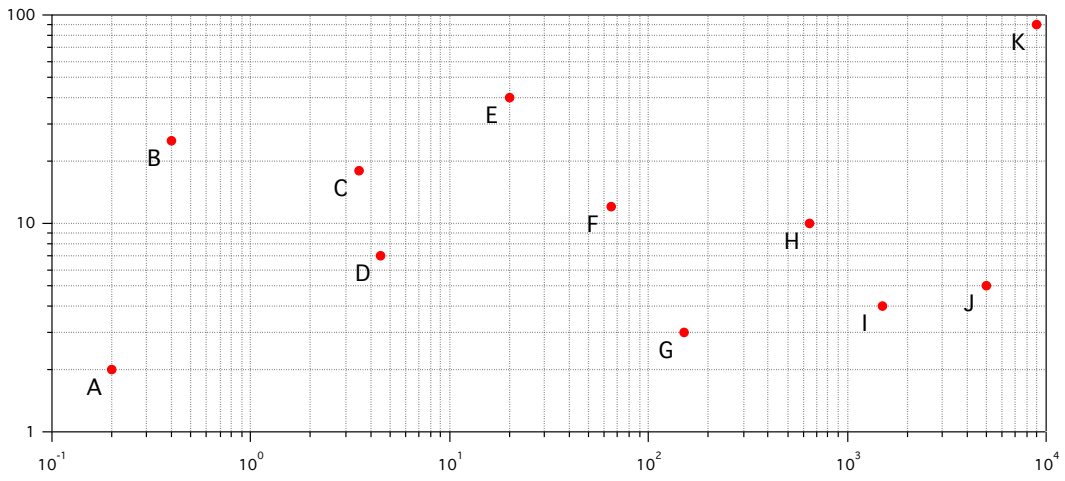
B: Years since the last ice age

Part 4: Log plots

1) Label all of the ticks on the log-log plot below



2) Record the values for the following data point on the table below. The first one has been done for you

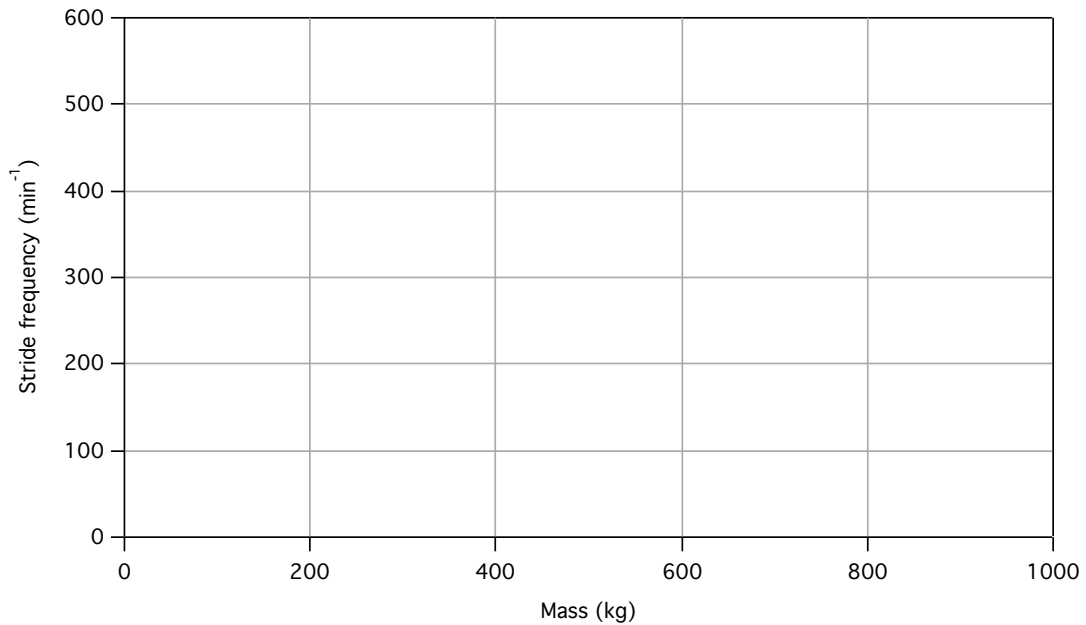


<i>Point</i>	<i>x-value</i>	<i>y-value</i>
A	0.2	2
B		
C		
D		
E		
F		
G		
H		
I		
J		
K		

- 3) Heglund, *et al.* (1974) measured the mass of several animals and their stride frequency at the transition from “trot” to “gallop”.[‡] The data they collected are shown in the Table below:

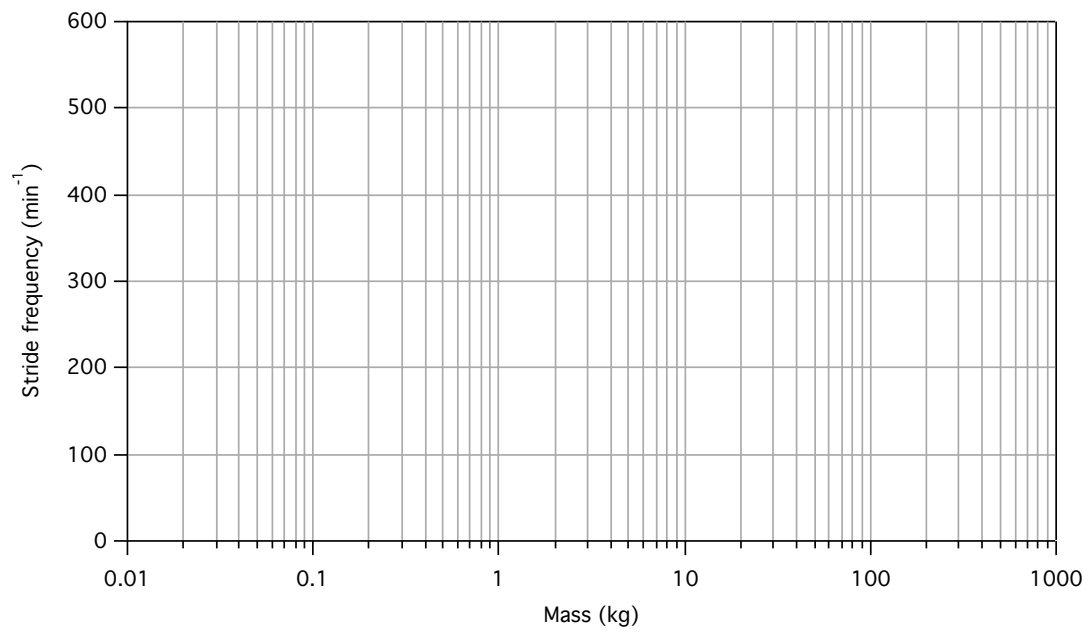
<i>Animal</i>	<i>Mass (kg)</i>	<i>Stride frequency (min⁻¹)</i>
Mouse	0.01	533
	0.03	444
	0.04	434
Rat	0.41	269
Dog	1.18	255
	9.65	198
	28.14	156
	30.91	153
Horse	700.4	117

Plot the data on a linear scale:



[‡] Heglund, N. C., Taylor, C. R., and McMahon, T. A. (1974). Scaling stride frequency and gait to animal size: Mice to horses. *Science*, 186, 1112-1113.

4) Plot the data on a semi-log scale:





Probability Practice Questions

- 1) On the first day of your Frontiers of Science (FoS) seminar, you choose your seat randomly and sit between two students you have never met before. What is the probability that the student sitting on your right was born on July 4th?
- 2) What is the probability that this student was born on July 4th OR February 14th?
- 3) What is the probability that this student was born on July 4th AND February 14th?
- 4) What is the probability that this student has the same birthday as you?
- 5) What is the probability that this student has the same birthday as you and is female?
- 6) What is the probability that both the student on your right and the student on your left have the same birthday as you?
- 7) What is the probability that at least one other student in your FoS seminar has the same birthday as you? (Assume that there are 20 students in your seminar.)
- 8) What is the probability that at least one other student currently taking FoS has the same birthday as you? (Assume that 500 students are currently taking FoS.)
- 9) Everyone in your seminar randomly picks a number between 1 and 50. Assuming your selections were truly random, is the probability that someone else has picked the same number as you?

Often the probability of an event occurring is estimated from the occurrence rate of that event in the past. The National Weather Service estimates that about 400 people in the U.S. were struck by lightning in 2011.⁵ The population of the U.S. in 2011 is estimated at 310 million.

- 10) From these data, estimate the probability of being struck by lightning in a given year. In other words, what is the occurrence rate of lightning strikes per person per year?
- 11) What is the probability (occurrence rate) of being struck by lightning in a given day?
- 12) What is the probability of being struck by lightning at least once in 10 days?
- 13) What is the probability of being struck by lightning at least once in a typical lifetime (80 years)?

⁵ <http://www.lightningsafety.noaa.gov/medical.htm>